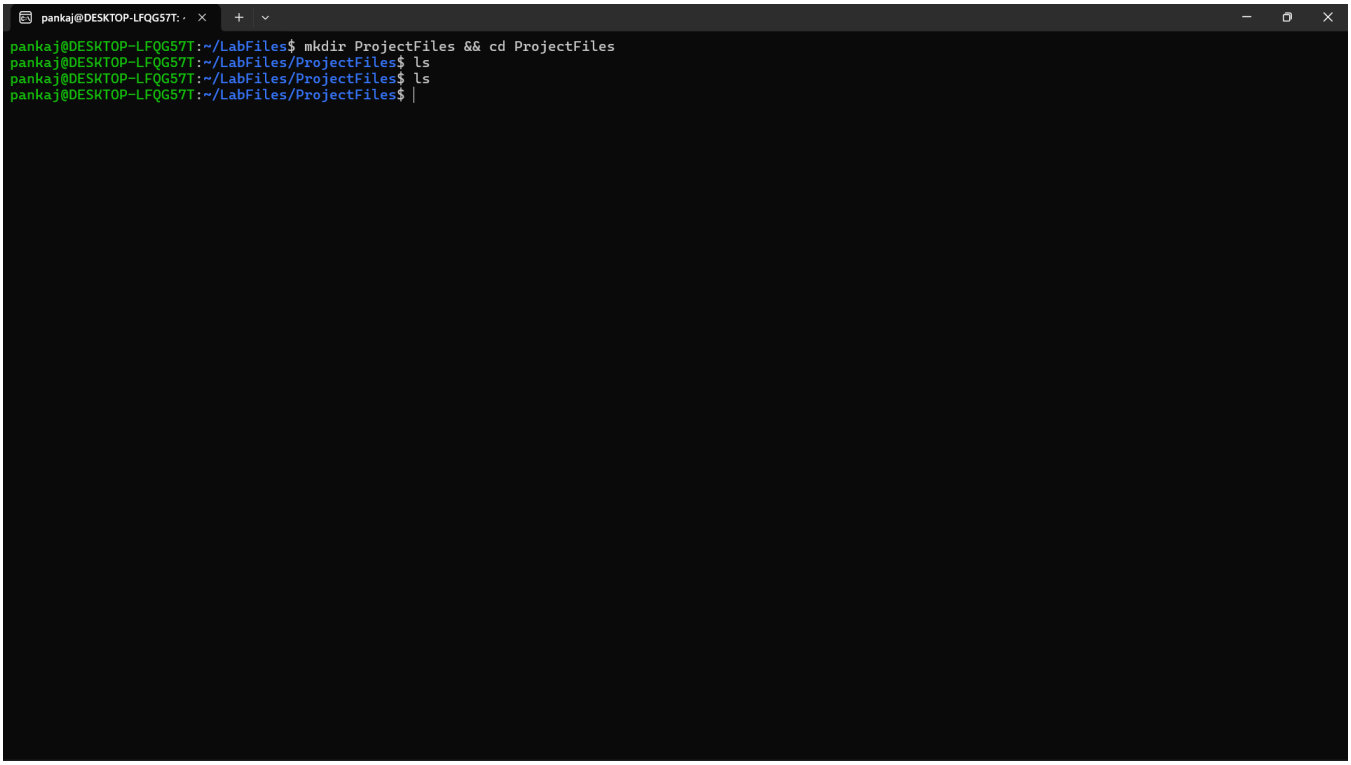
# Question 5

**Configure file permissions & ownership (symbolic/numeric chmod, chown, chgrp, umask) and verify with ls -l, including recursive updates**

## Step 1 – Create the ProjectFiles directory

Commands used: mkdir ProjectFiles && cd ProjectFiles, ls.

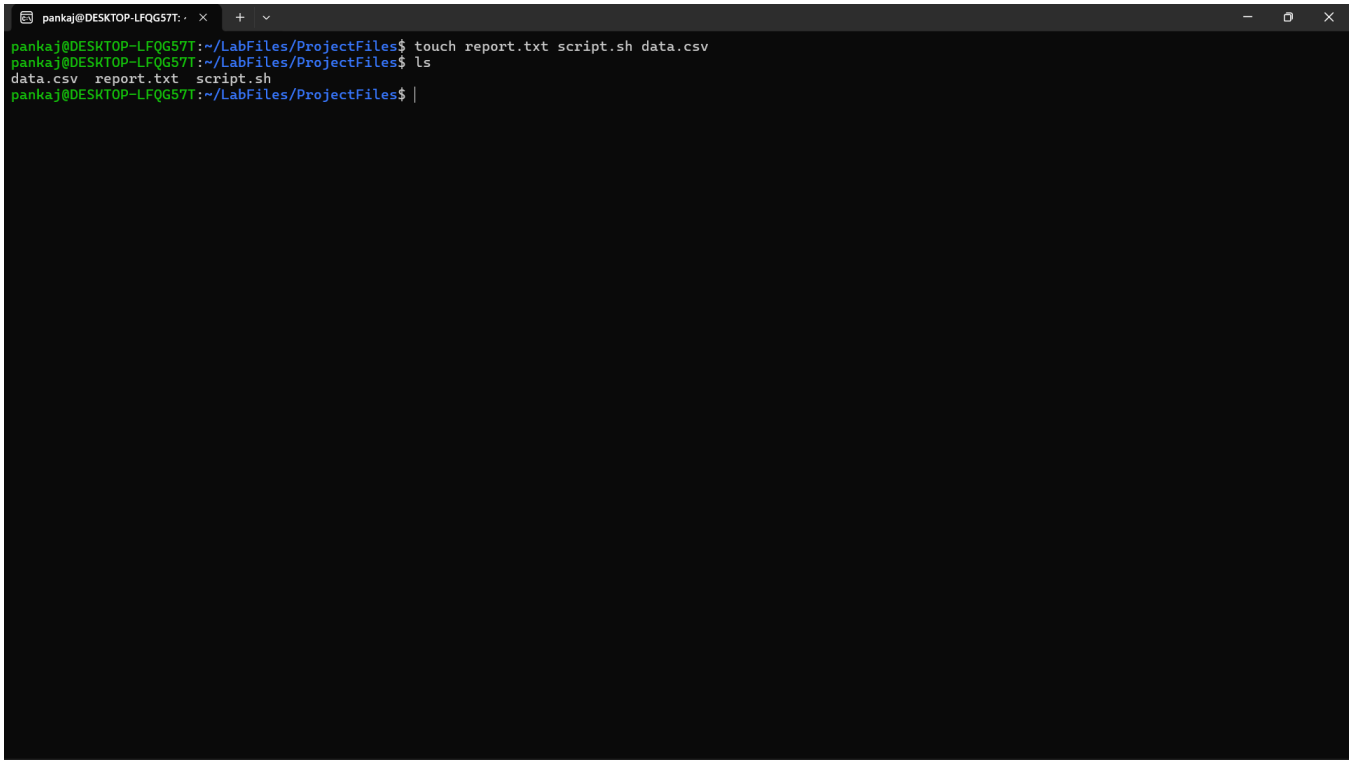
We created the required ProjectFiles directory and entered it. ls confirmed the directory was empty and ready for the sample files.



## Step 2 – Create the sample files

Commands used: touch report.txt script.sh data.csv, ls.

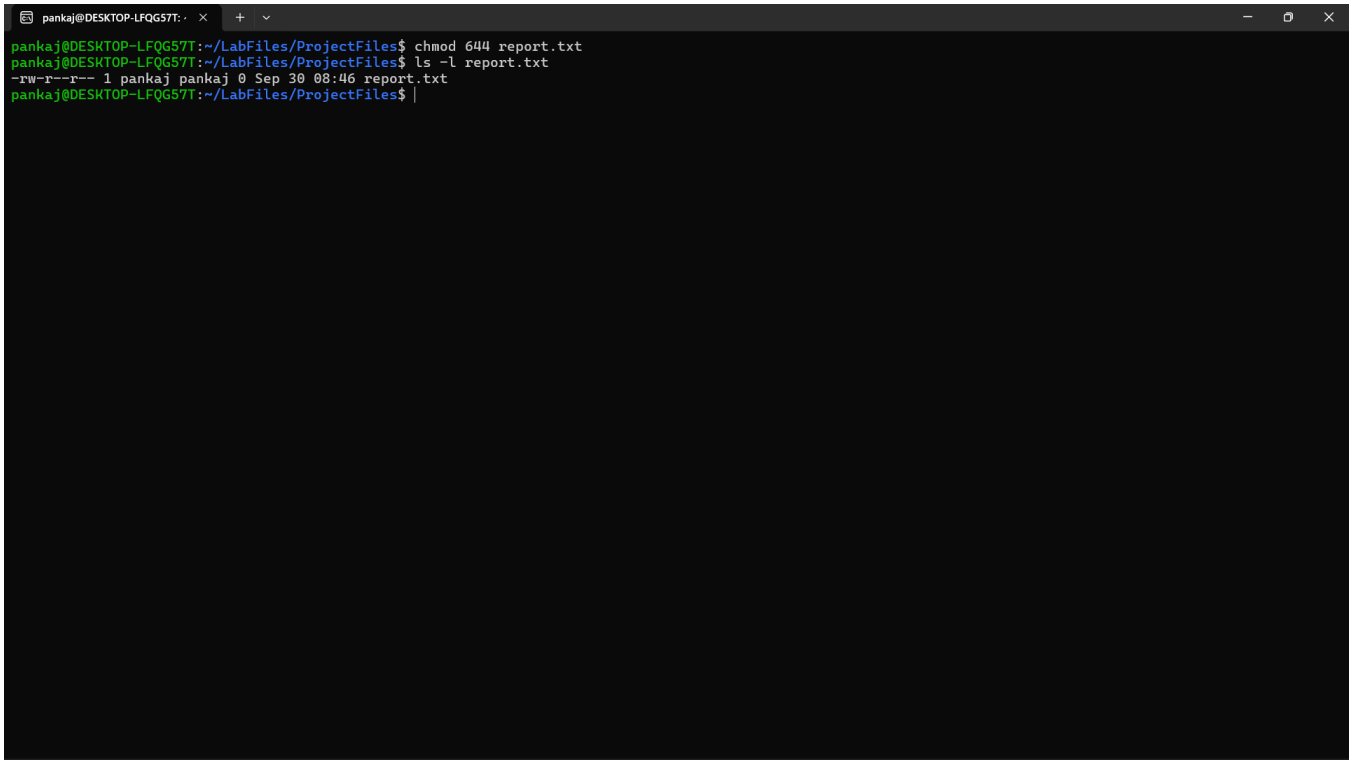
We created report.txt, script.sh and data.csv using touch. ls confirmed that all three files were present.




## Step 3 – Apply numeric permission 644

Commands used: chmod 644 report.txt, ls -l report.txt.

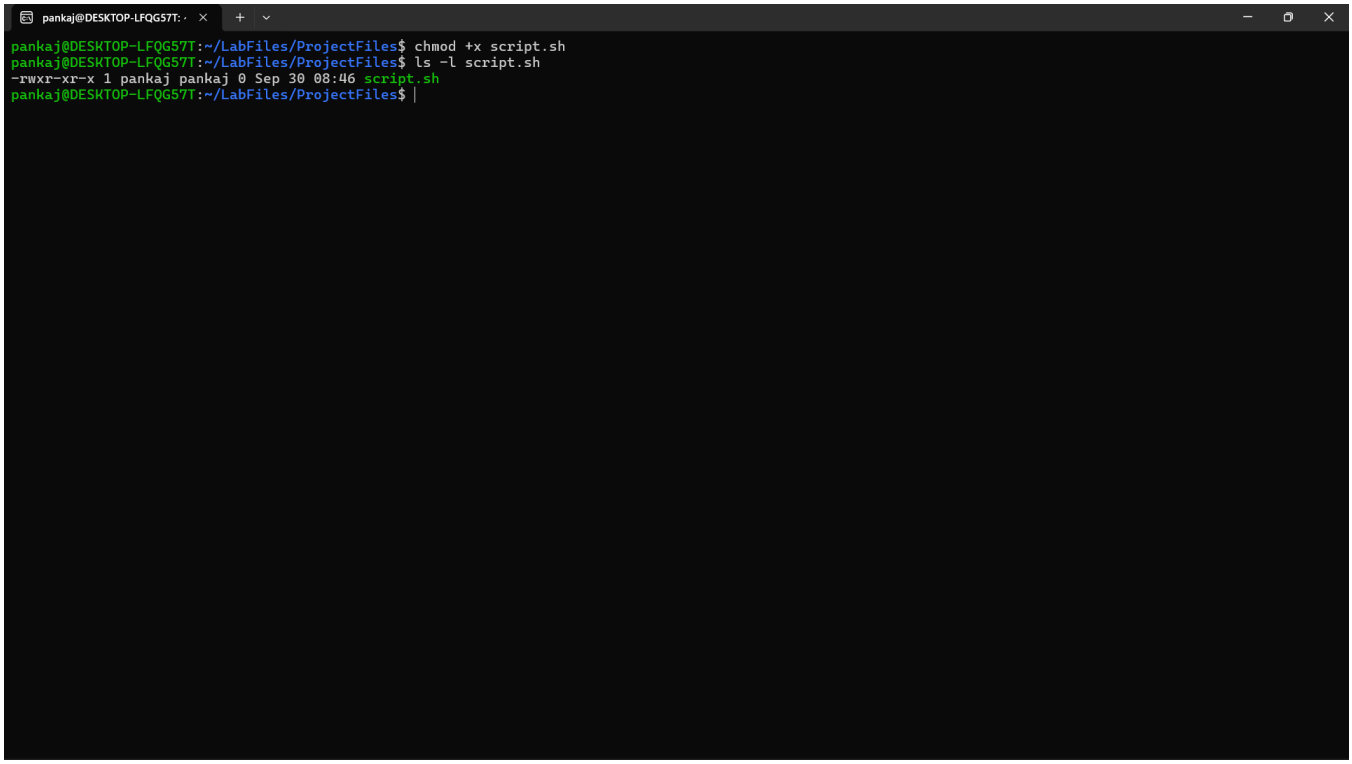
We applied numeric permission 644 to report.txt and verified the resulting read/write permissions with ls -l.




## Step 4 – Apply symbolic execute permission

Commands used: chmod +x script.sh, ls -l script.sh.

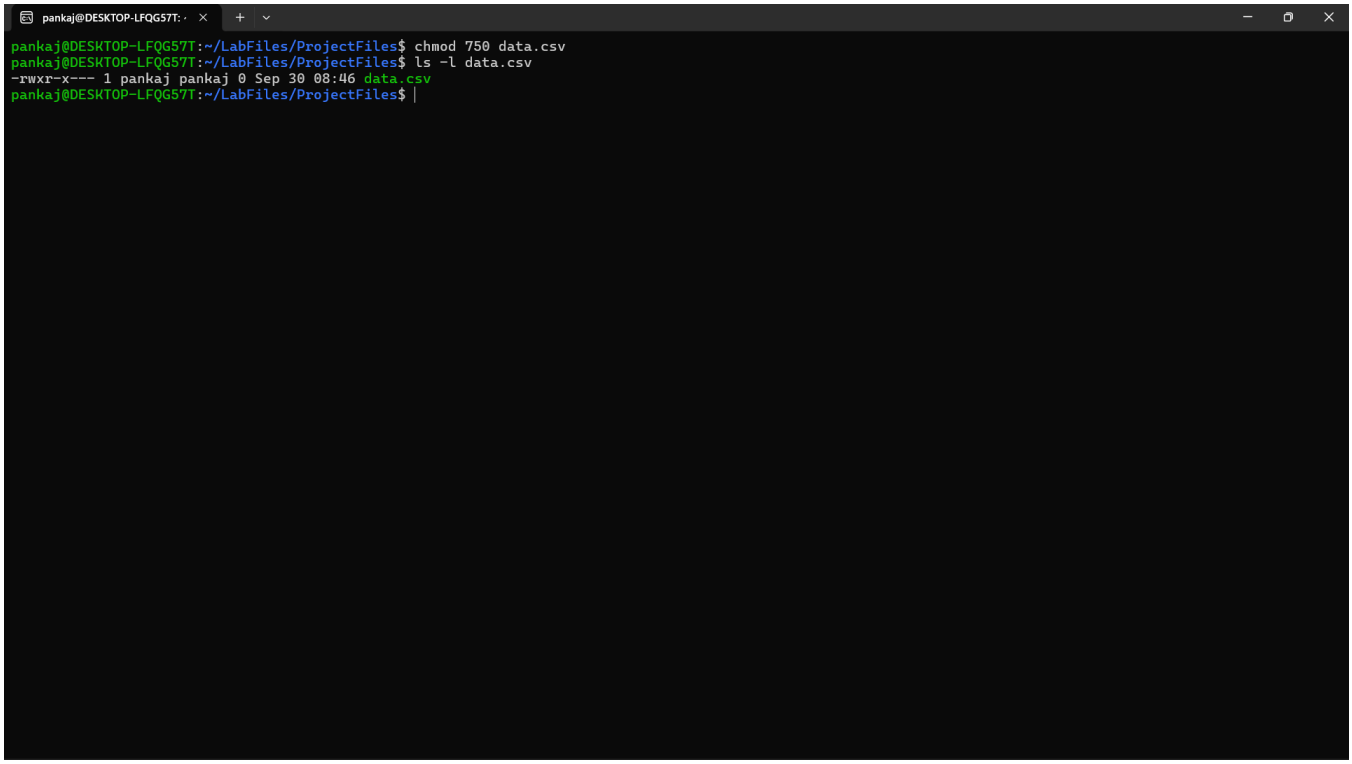
We used symbolic chmod to add execute permission to script.sh and verified the resulting permission string.




## Step 5 – Apply numeric permission 750

Commands used: chmod 750 data.csv, ls -l data.csv.

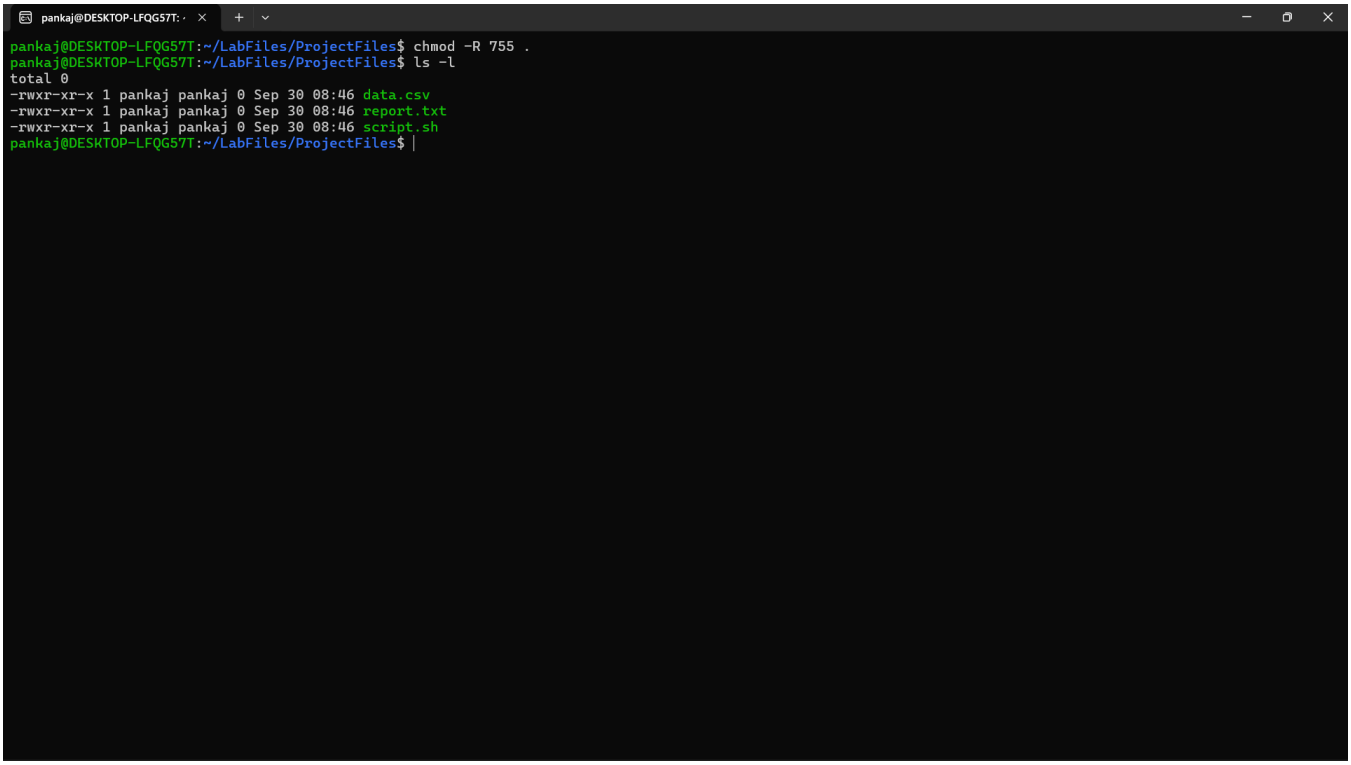
We applied numeric permission 750 to data.csv. The listing shows -rwxr-x---, which verifies the required 750 permission.




## Step 6 – Apply permissions recursively

Commands used: chmod -R 755 ., ls -l.

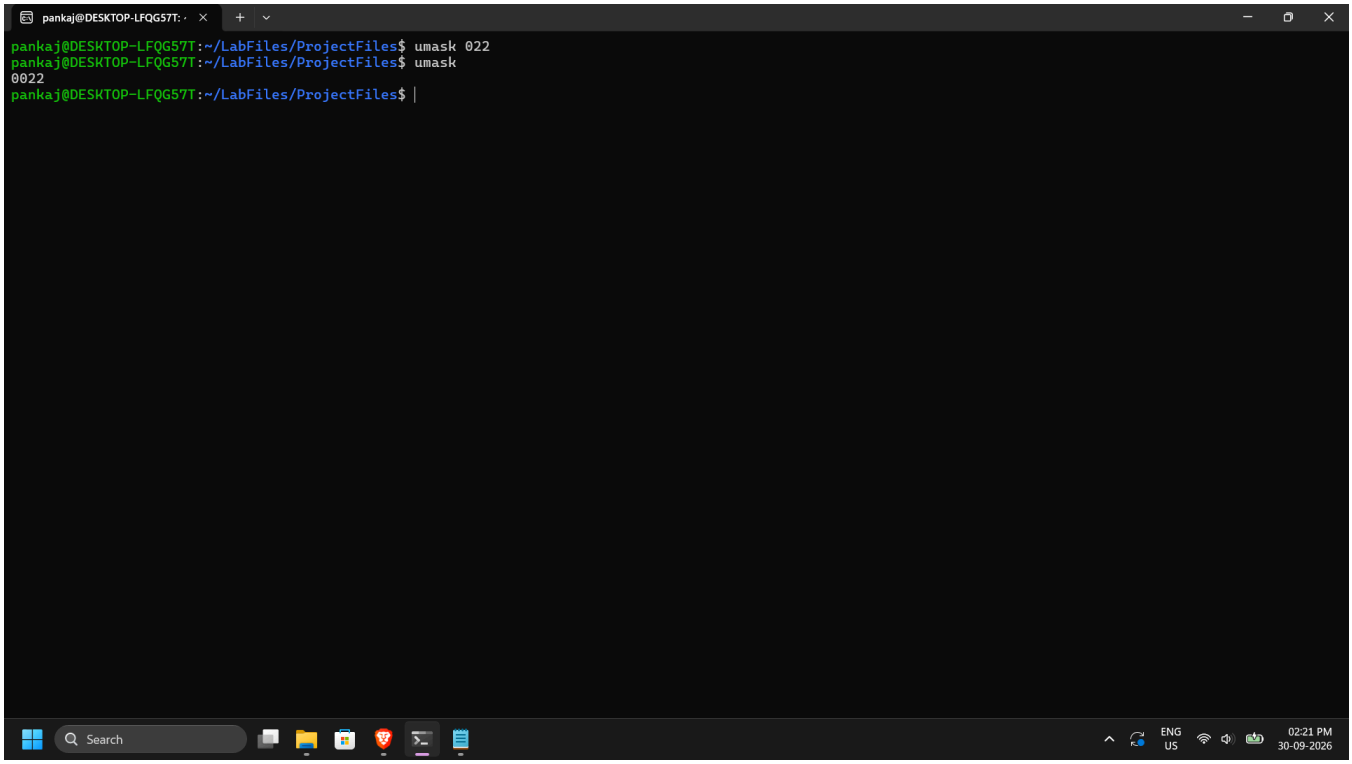
We applied 755 recursively to the current ProjectFiles directory and its contents. The long listing confirms the updated permissions on the files.




## Step 7 – Set and verify the default umask

Commands used: umask 022, umask.

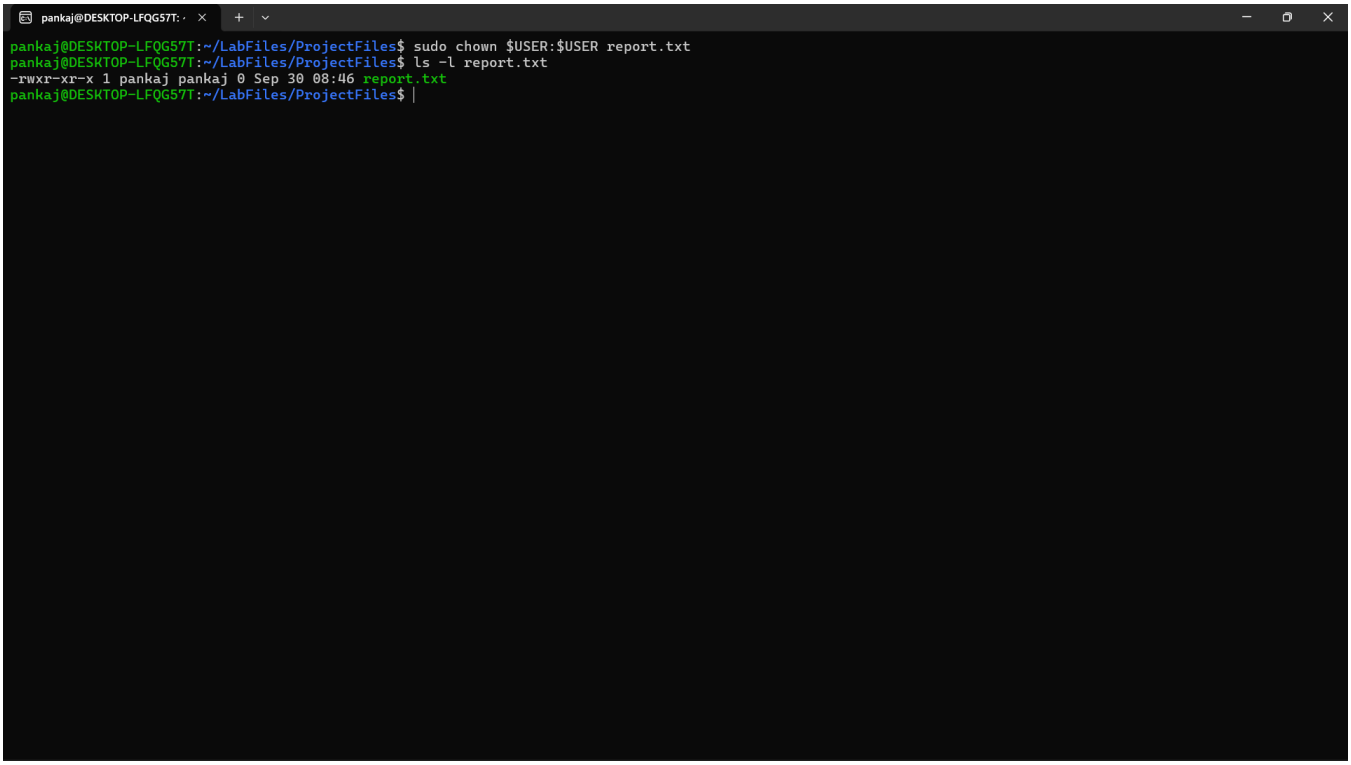
We set the default creation mask to 022 and then used umask to verify that the active mask is 0022.




## Step 8 – Change file ownership

Commands used: sudo chown $USER:$USER report.txt, ls -l report.txt.

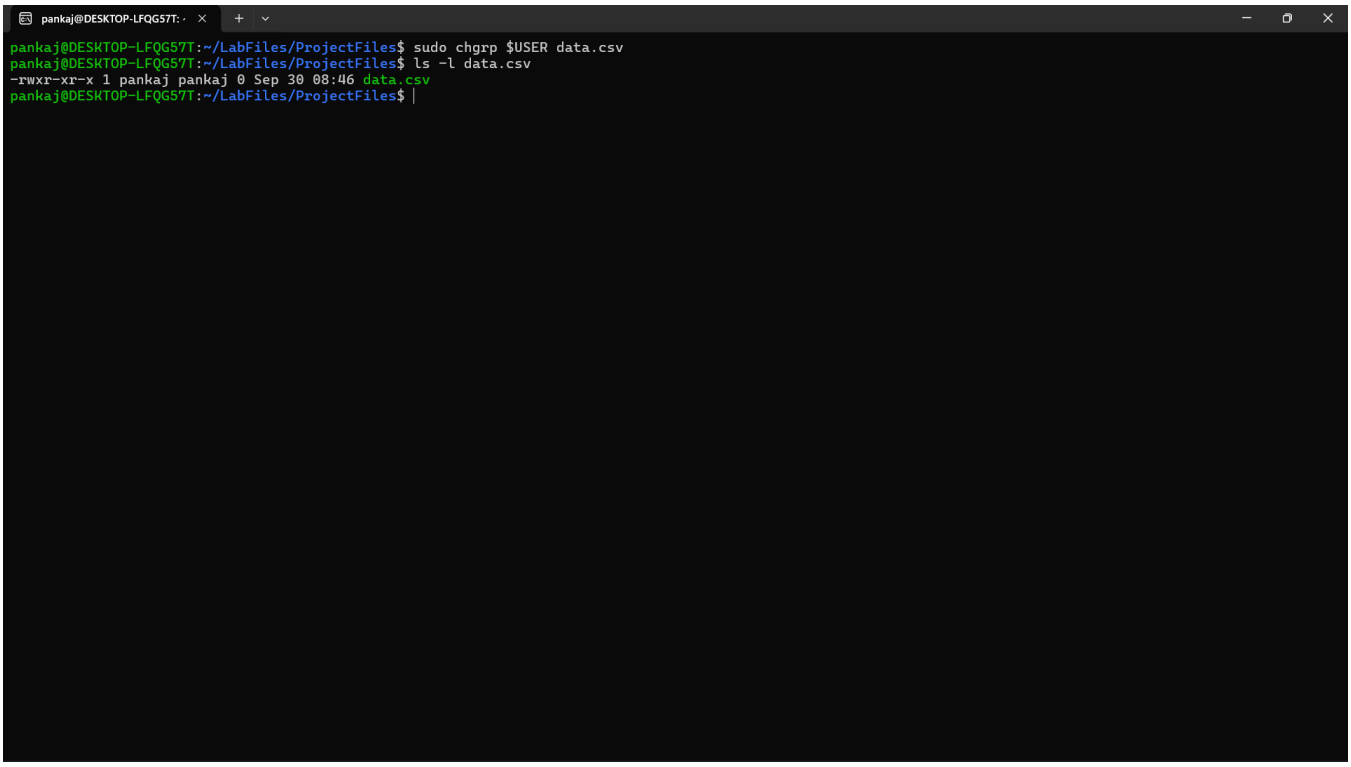
We changed the owning user and group of report.txt using the current user represented by $USER, then verified the ownership with ls -l.




## Step 9 – Change the file group

Commands used: sudo chgrp $USER data.csv, ls -l data.csv.

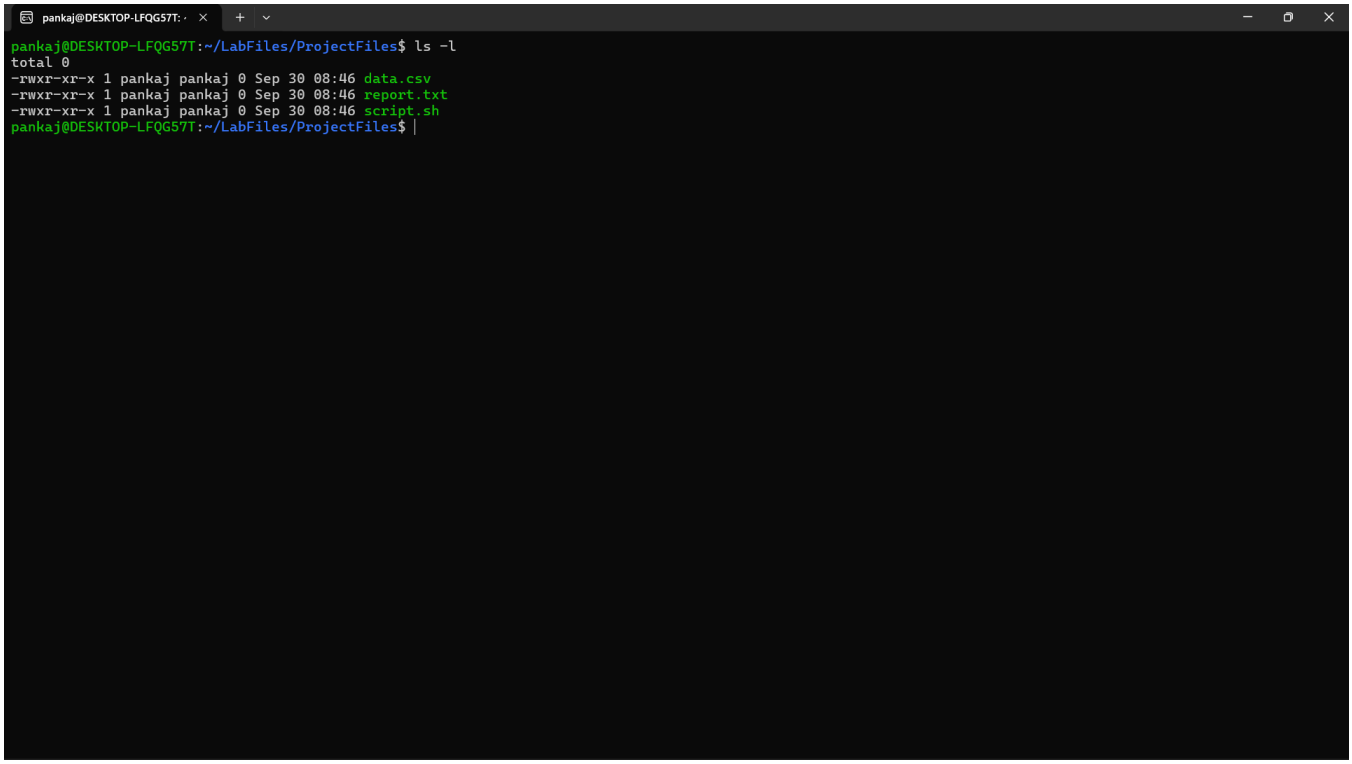
We changed the owning group of data.csv using $USER and verified the result with ls -l.




## Step 10 – Verify the final permissions and ownership

Commands used: ls -l.

We displayed the final long listing of the ProjectFiles directory to verify the permissions, owner and group of all three files.


pankaj@DESKTOP-LFQG57T:~/LabFiles/ProjectFiles$ ls -l
total 0
-rwxr-xr-x 1 pankaj pankaj 0 Sep 30 08:46 data.csv
-rwxr-xr-x 1 pankaj pankaj 0 Sep 30 08:46 report.txt
-rwxr-xr-x 1 pankaj pankaj 0 Sep 30 08:46 script.sh
pankaj@DESKTOP-LFQG57T:~/LabFiles/ProjectFiles$